

Mock Test — Nature of Matter: Elements, Compounds, and Mixtures

A mixed paper covering the whole chapter. Total: 20 marks. Section A is interactive.

Objective (1 mark each)

Q1 • MCQ

[1]

In a mixture, the components —

- (a) keep their own properties (b) form new properties (c) react chemically
(d) are in a fixed ratio

Q2 • MCQ

[1]

Which is an element?

- (a) water (b) air (c) oxygen (d) brass

Q3 • MCQ

[1]

Brass is an alloy of copper and —

- (a) tin (b) zinc (c) iron (d) nickel

Q4 • MCQ

[1]

Water is a —

- (a) mixture (b) compound (c) element (d) alloy

Q5 • MCQ

[1]

The gas that burns with a pop is —

- (a) oxygen (b) hydrogen (c) nitrogen (d) argon

Q6 · TRUE / FALSE

[1]

A compound can be separated into elements by filtering.

(a) True (b) False

Q7 · FILL IN THE BLANK

[1]

A uniform mixture of metals is an _____.

Q8 · FILL IN THE BLANK

[1]

Lime water turns _____ in the presence of carbon dioxide.

Short answer (2 marks each)

Q9

[2]

Differentiate between an element and a compound with one example each.

Q10

[2]

Give two differences between a mixture and a compound.

Q11

[2]

Why is air a mixture and water a compound?

Long / application (3 marks each)

Q12

[3]

Classify and justify: oxygen, air, water, brass — as element, compound or mixture.

Q13

[3]

Describe the electrolysis of water and state how the products prove water is a compound.

[Check yourself](#) → [Answer key](#)